



ABHAY MAHAJAN

B.Tech. IIT Roorkee

- 11+ years teaching experience
- Mentored and taught around 20000+ students
- Produced 400+ IIT selections
- 10+ INMO selections
- 1 EGMO Selection



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KVPY SA SCHEDULE – WEEK 1

**ABHAY
MAHAJAN**

Number Theory

**MONDAY
@ 11:00 AM**

**NISHANT
SHARMA**

**Kinematics and Laws of Motion:
Most Important Questions**

**WEDNESDAY
@ 12:30 PM**

**ASIF
TYAGI**

**Animal Tissue:
Part-1**

**WEDNESDAY
@ 4:00 PM**

**NIDHI
PRASHAR**

Atomic Structure

**FRIDAY
@ 11:30 AM**

KVPY SX SCHEDULE – WEEK 1

**ABHAY
MAHAJAN**

**Sequence and Series
and Quadratic Equations**

**WEDNESDAY
@ 11:00 AM**

**NISHANT
SHARMA**

**Kinematics and Laws of Motion:
Most Important Questions**

**THURSDAY
@ 12:30 PM**

**ASIF
TYAGI**

**Animal Reproduction:
Part-1**

**FRIDAY
@ 4:00 PM**

**NIDHI
PRASHAR**

Solutions and Colligative Properties

**SATURDAY
@ 11:30 AM**



Only on
VOS Youtube Channel

EVERY TUESDAY

PRE-RMO

4 PM

EVERY THURSDAY

PRE-RMO

4 PM

EVERY SATURDAY

PRE-RMO

4 PM

ABHAY MAHAJAN
IIT ROORKEE



For
DOUBTS , QUERIES & MOTIVATION
Reach out to me...



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ABHAY MAHAJAN
MATHEMATICS EXPERT

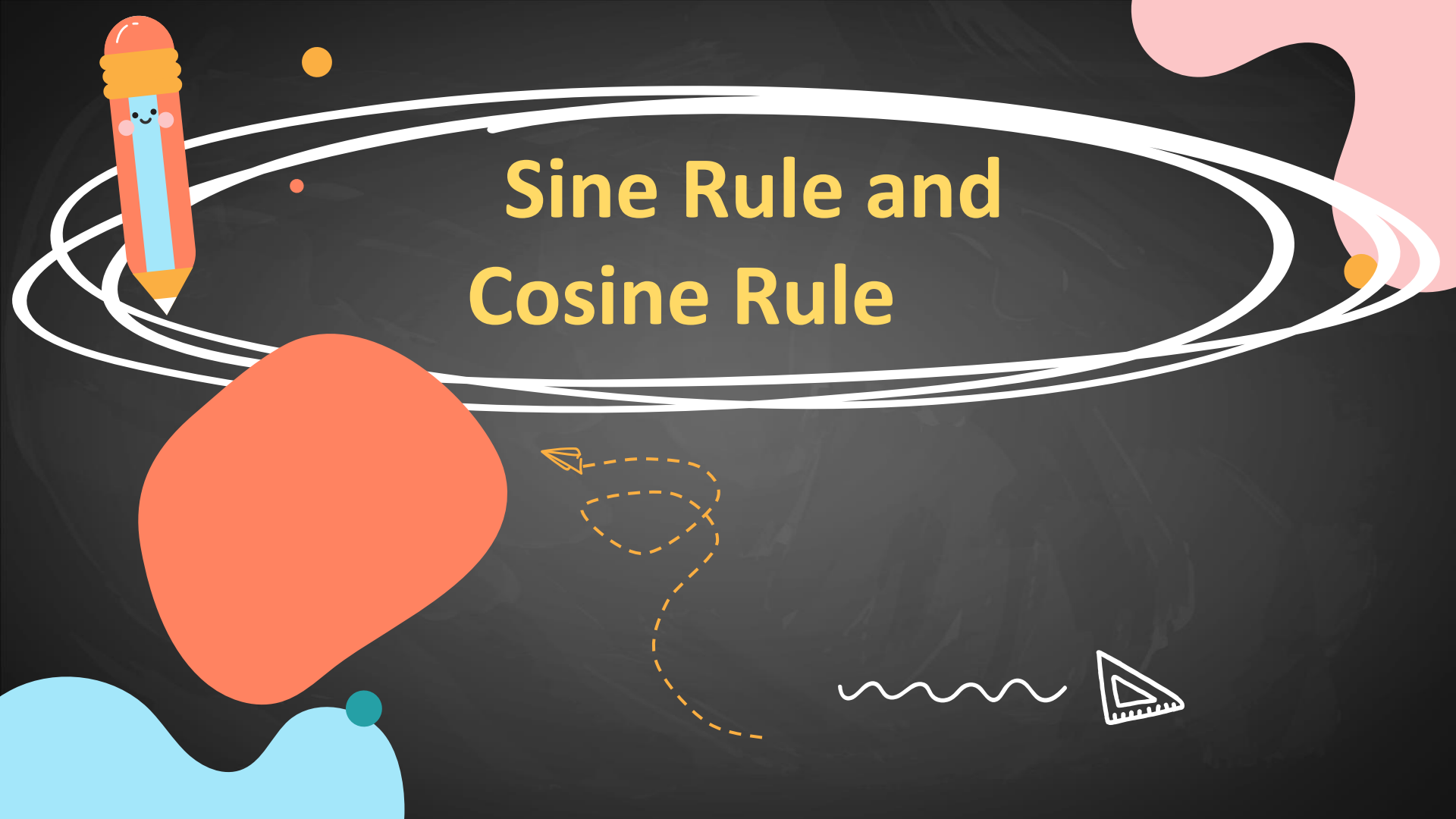


Content Creator :

Sahil Jindal

B.Tech, NIT Kurukshetra

4 Year JEE Teaching
Experience.

The background is a dark grey chalkboard. A large, hand-drawn white oval frame encircles the title. To the left of the frame, a cartoon pencil with a face and a blue band is positioned vertically. Several white orbital lines swirl around the pencil. Large, soft-edged shapes in orange, light blue, and pink are scattered around the board. A dashed orange line with an arrowhead is in the lower center, and a white wavy line and a set square are in the lower right.

Sine Rule and Cosine Rule

Sine Rule

In any $\triangle ABC$,

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = \frac{1}{2R}$$

$$a : b : c = \sin A : \sin B : \sin C$$

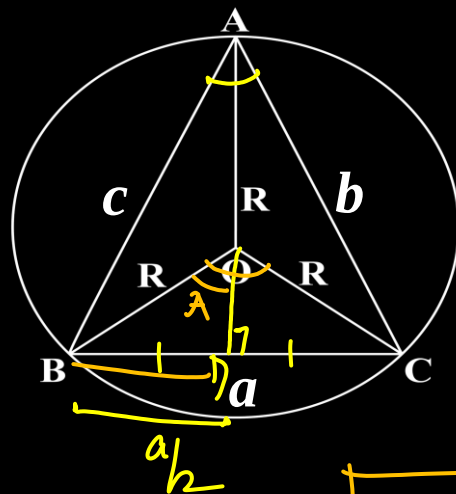
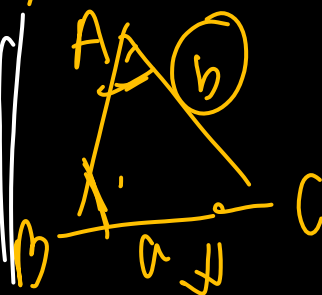
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

Relation b/w sides & angles

$$a = 2R \sin A$$

$$b = 2R \sin B$$

$$c = 2R \sin C$$



$\triangle OBD$

$$\frac{BD}{OB} = \sin A$$

$$\frac{a/2}{R} = \sin A$$

$$\frac{a}{\sin A} = 2R$$

Cosine Rule

$$\cos C = \frac{CD}{b}$$

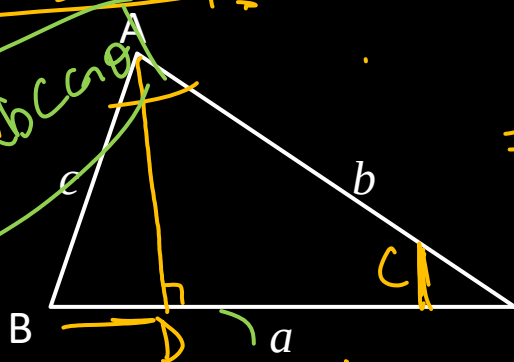
In any $\triangle ABC$,

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$CD = b \cos C$$



$$\begin{aligned} \triangle ADC: \quad b^2 &= AD^2 + CD^2 \\ &= x^2 + CD^2 \\ &= x^2 + (a-x)^2 + CD^2 \\ &= x^2 + (a^2 + x^2 - 2ax) + CD^2 \\ &= 2x^2 - 2ax + a^2 + CD^2 \end{aligned}$$

$$\begin{aligned} b^2 + c^2 - a^2 &= 2bc \cos A \\ \checkmark \quad a^2 &= b^2 + c^2 - 2bc \cos A \end{aligned}$$

$$b^2 = c^2 - a^2 + 2a(CD) + b \cos C$$

$$2ab \cos C = a^2 + b^2 - c^2$$

Q1. In $\triangle ABC$, if $(a + b + c)(a - b + c) = 3ac$, then find $\angle B$ (in degrees) .

Cute
Question

$$(a+c)^2 - b^2 = 3ac$$

$$\frac{a^2 + c^2 - b^2}{2ac} = \frac{ac}{2ac}$$

$$\cos B = \frac{1}{2}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\angle B = 60^\circ \checkmark$$

Q1. In $\triangle ABC$, if $(a + b + c)(a - b + c) = 3ac$, then find $\angle B$ (in degrees) .

Answer: 60°

Solution:

We have $(a + c)^2 - b^2 = 3ac$

Or $a^2 + c^2 - b^2 = ac$

But $\cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{1}{2}$

So, $B = 60^\circ$

$$a = 2R \sin A$$

Q2. The perimeter of a triangle ABC is six times the arithmetic mean of the sines of its angles. If the side a is 1, then find the angle A in degrees.

Cute Ques

$$a+b+c = 6 \left(\frac{\sin A + \sin B + \sin C}{3} \right),$$

$$\boxed{R=1}$$

$$A=?$$

$$30^\circ, 150^\circ$$

$$2R(\sin A + \sin B + \sin C) = 2(\sin A + \sin B + \sin C)$$

$$\boxed{R=1}$$

$$\frac{a}{\sin A} = 2R, \quad \sin A = \frac{a}{2R} = \frac{1}{2 \cdot 1} = \frac{1}{2}$$

Q2. The perimeter of a triangle ABC is six times the arithmetic mean of the sines of its angles. If the side a is 1, then find the angle A in degrees.

Answer: 30°

Solution:

$$\text{Given that } a + b + c = 6 \times \frac{\sin A + \sin B + \sin C}{3}$$

$$\text{or } 2R(\sin A + \sin B + \sin C) = 2(\sin A + \sin B + \sin C)$$

$$\Rightarrow R = 1$$

$$\text{Now, } \frac{a}{\sin A} = 2R$$

$$\text{or } \sin A = \frac{1}{2} (\because a = 1) \Rightarrow A = 30^\circ$$

Q3. In a triangle ABC, if the sides a, b, c are roots of

$x^3 - 11x^2 + 38x - 40 = 0$, then find the value of

$$16\left(\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c}\right).$$

$$\begin{array}{l} a \rightarrow A \\ b \rightarrow B \\ c \rightarrow C \end{array}$$

9

$$16 \left[\frac{b^2 + c^2 - a^2}{2bc} \cdot \frac{1}{a} + \frac{a^2 + c^2 - b^2}{2ac} \cdot \frac{1}{b} + \frac{a^2 + b^2 - c^2}{2ab} \cdot \frac{1}{c} \right]$$

$$\frac{16}{2abc} [a^2 + b^2 + c^2] = \frac{8}{abc} [(a+b+c)^2 - 2(ab+bc+ca)] = \frac{8}{40} [11^2 - 2 \cdot 38]$$

$$a+b+c=11; ab+bc+ca=38$$

$$abc=40$$

$$= \frac{1}{5} [121 - 76] = \frac{45}{5}$$

Q3. In a triangle ABC, if the sides a , b , c are roots of $x^3 - 11x^2 + 38x - 40 = 0$, then find the value of $16\left(\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c}\right)$.

Answer: 9

Solution:

Here a, b, c are roots of equation $x^3 - 11x^2 + 38x - 40 = 0$.

Therefore,

$a + b + c = 11$, $ab + bc + ac = 38$, and $abc = 40$

$$\Rightarrow \frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a^2 + b^2 + c^2}{2abc}$$

$$= \frac{(a + b + c)^2 - 2(ab + bc + ac)}{2abc}$$

$$= \frac{11^2 - 76}{80} = \frac{45}{80} = \frac{9}{16}$$

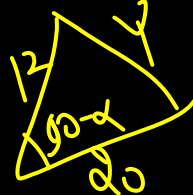
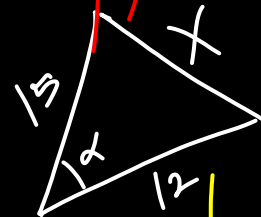
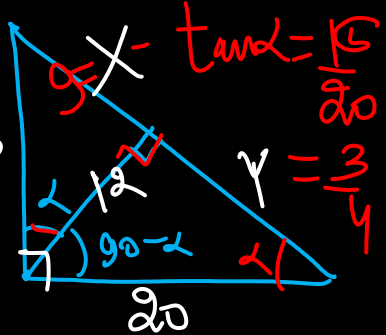
Q4. Given that α is an acute angle satisfying

$$\sqrt{369 - 360 \cos \alpha} + \sqrt{544 - 480 \sin \alpha} - 25 = 0$$

find the value of $40 \tan \alpha$.

$$20^2 + 15^2 = 25^2$$

$$15 \cdot 20 = 25 \cdot 12$$



$$\cos(90-\alpha)$$

$$Y^2 = 544 - 480 \sin \alpha$$

$$= c^2 + d^2 - 2cd \sin \alpha$$

$$c^2 + d^2 = 544 = 20^2 + 12^2 - 2 \cdot 20 \cdot 12$$

$$2cd = 480, cd = 240 \quad \cos(90-\alpha)$$

$$20, 12$$

$$X^2 = 15^2 + 12^2 - 2 \cdot 15 \cdot 12 \cos \alpha$$

$$(a+b)^2 = a^2 + b^2 + 2ab$$

$$15^2 + 12^2 + 2 \cdot 15 \cdot 12 = 25^2$$

$$225 + 144 + 360 = 625$$

$$625 = 625$$

$$(a-b)^2 = a^2 + b^2 - 2ab$$

$$15 \cdot 12$$

$$ab = 180$$

$$a^2 + b^2 = 369$$

$$2ab = 360$$

$$X^2 = 369 - 360 \cos \alpha$$

$$X^2 = a^2 + b^2 - 2ab \cos \alpha$$

$$X + Y = 25$$

$$(a+b)^2$$

$$a^2 + b^2 + 2ab$$

$$15^2 + 12^2 + 2 \cdot 15 \cdot 12$$

$$225 + 144 + 360$$

$$625 = 25^2$$

$$(a-b)^2 = a^2 + b^2 - 2ab$$

$$15^2 + 12^2 - 2 \cdot 15 \cdot 12$$

$$225 + 144 - 360$$

$$625 - 360 = 265$$

Q4. Given that α is an acute angle satisfying

$$\sqrt{369 - 360 \cos \alpha} + \sqrt{544 - 480 \sin \alpha} - 25 = 0$$

find the value of $40 \tan \alpha$.

Answer: 30

Solution:

Let $X = \sqrt{369 - 360 \cos \alpha}$ and $Y = \sqrt{544 - 480 \sin \alpha}$.

Observe that

$$X^2 = 12^2 + 15^2 - 2(12)(15)\cos \alpha$$

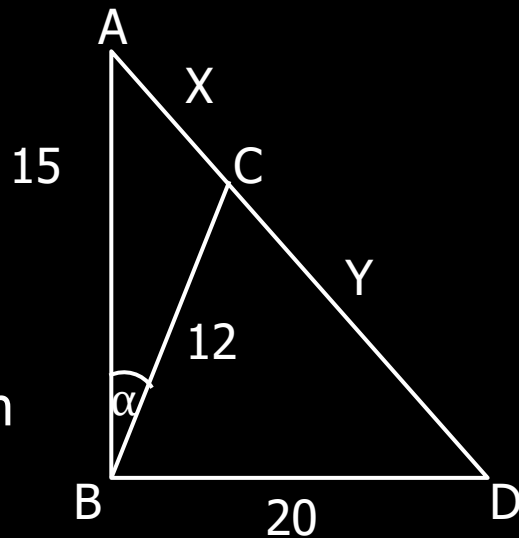
$$Y^2 = 12^2 + 20^2 - 2(12)(20)\cos(90^\circ - \alpha)$$

and $15^2 + 20^2 = 25^2 = (X + Y)^2$, so we can construct a right-angled triangle ABD as shown.

In particular $\angle ABC = \alpha$ and $\angle CBD = 90^\circ - \alpha$.

We can check that $\triangle ACB$ is in fact similar to $\triangle ABD$. So $\angle ADB = \alpha$ and

$$40 \tan \alpha = 40 \times \frac{15}{20} = 30$$



Q

JEE KVPY

Q5. In a triangle ABC, $\angle C = 60^\circ$ and $\angle A = 75^\circ$. If D is a point on AC such that the area of the $\triangle ABD$ is $\sqrt{3}$ times the area of the $\triangle BCD$, find the $\angle ABD$.

$$\sin 75 = \sin(45 + 30)$$

$$\frac{\sin 60}{\sin 75} = \frac{\sqrt{3} \sin(45 - \theta)}{\sin \theta \cdot 1}$$

$$\frac{\sqrt{3} \sin \theta}{\sqrt{3} + 1} = \sqrt{3} \left(\frac{1}{\sqrt{2}} \cos \theta - \frac{1}{\sqrt{2}} \sin \theta \right)$$

$$2 \sin \theta = (\sqrt{3} + 1)(\cos \theta - \sin \theta)$$

$$\sqrt{3}(\sqrt{3} + 1) \sin \theta = (\sqrt{3} + 3) \sin \theta = (\sqrt{3} + 1) \cos \theta$$

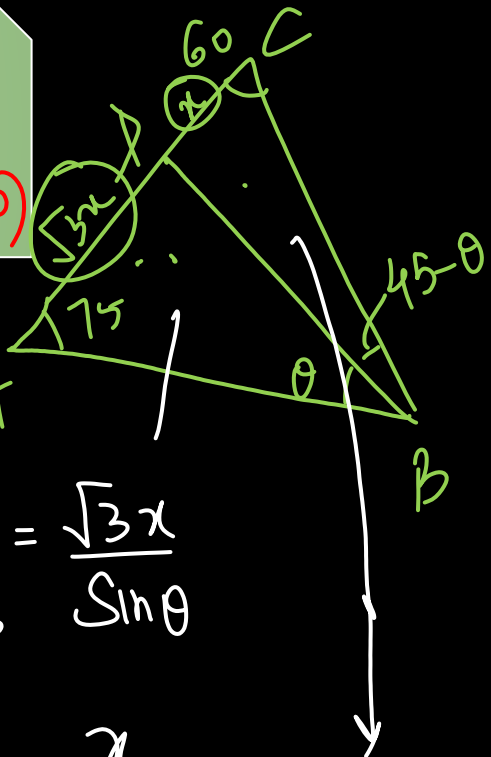
$$\sqrt{3} \sin \theta = \cos \theta$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\theta = 30$$

$$\frac{BD}{\sin 75} = \frac{\sqrt{3} x}{\sin \theta}$$

$$\frac{BD}{\sin 60} = \frac{x}{\sin(45 - \theta)}$$



Q5. In a triangle ABC, $\angle C = 60^\circ$ and $\angle A = 75^\circ$. If D is a point on AC such that the area of the $\triangle ABD$ is $\sqrt{3}$ times the area of the $\triangle BCD$, find the $\angle ABD$.

Answer: 30°

Solution:

Let h be the length of perpendicular from B on AC

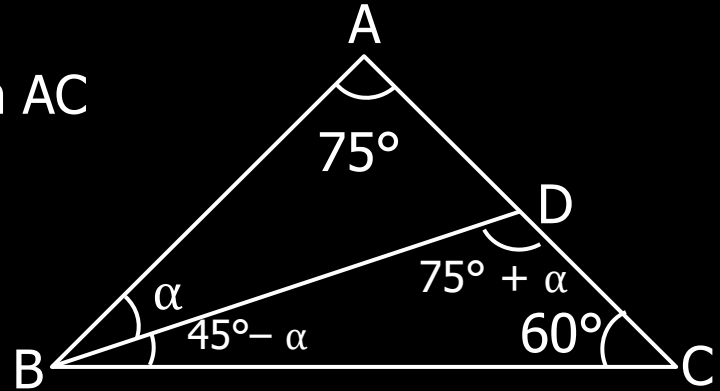
Given that $\frac{\Delta BAD}{\Delta BCD} = \sqrt{3}$

$$\Rightarrow \frac{\frac{1}{2}h \cdot AD}{\frac{1}{2}h \cdot DC} = \frac{AD}{CD} = \sqrt{3} \quad \dots\dots(1)$$

In ΔBAD , taking $\angle ABD = \alpha$ we have

$$\frac{AD}{\sin \alpha} = \frac{BD}{\sin 75^\circ} \quad \dots\dots(2)$$

And in ΔBCD , we have $\frac{CD}{\sin(45^\circ - \alpha)} = \frac{BD}{\sin 60^\circ} \quad \dots\dots(3)$



Solution:

∴ From (2) and (3), we get

$$\frac{AD \sin(45^\circ - \alpha)}{CD \sin \alpha} = \frac{\sin 60^\circ}{\sin 75^\circ}$$

$$\Rightarrow \sqrt{3} \sin 75^\circ \sin(45^\circ - \alpha) = \sin 60^\circ \cdot \sin \alpha$$

$$\Rightarrow \sqrt{3} \left(\frac{\sqrt{3} + 1}{2\sqrt{2}} \right) \left(\frac{\cos \alpha - \sin \alpha}{\sqrt{2}} \right) = \frac{\sqrt{3}}{2} \sin \alpha$$

$$\Rightarrow (\sqrt{3} + 1) \cos \alpha = (3 + \sqrt{3}) \sin \alpha$$

$$\Rightarrow \tan \alpha = \frac{1}{\sqrt{3}}$$

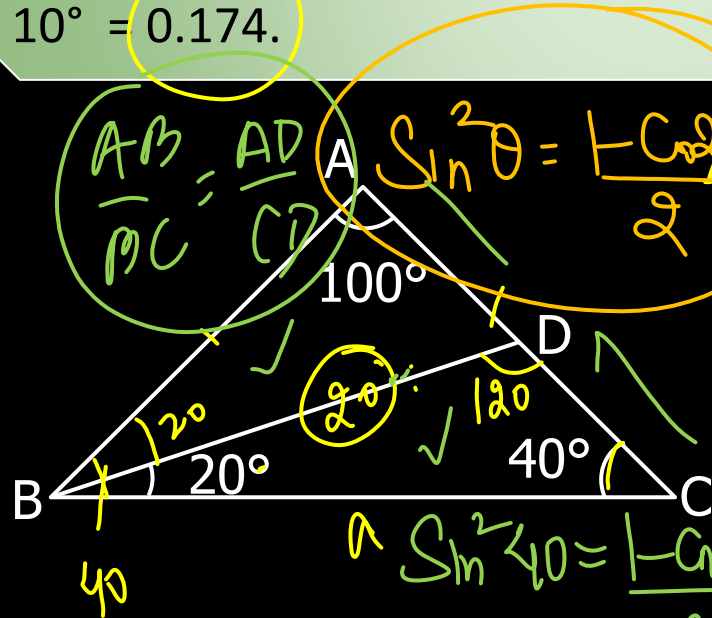
$$\Rightarrow \alpha = \frac{\pi}{6} = 30^\circ$$

Hence $\angle ABD = 30^\circ$

Thursday: Trigonometry for PRMO

PRMO - 2019

Q6. In a triangle ABC, it is known that $\angle A = 100^\circ$ and $AB = AC$. The internal angle bisector BD has length 20 units. Find the length of BC to the nearest integer, given that $\sin 10^\circ = 0.174$.



$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}; \quad \frac{a}{\sin 120} = \frac{20}{\sin 40}$$

$$a = \frac{20 \sin 120}{\sin 40} = \frac{20 \cdot \sin 60}{\sin 40}$$

$$a \sin^2 40 = \frac{1 - \cos 80}{2} = \frac{1 - 0.174}{2} = \frac{0.826}{2} = 0.413$$

$$\begin{aligned} \checkmark \quad \sin 10^\circ &= 0.174 \\ \text{Cn 80} &= \frac{20 \sqrt{3}}{2 \sin 40} \\ &= \frac{10 \sqrt{3}}{\sin 40} = \frac{10 \times 1.73}{\sqrt{0.413}} \end{aligned}$$

Q6. In a triangle ABC, it is known that $\angle A = 100^\circ$ and $AB = AC$. The internal angle bisector BD has length 20 units. Find the length of BC to the nearest integer, given that $\sin 10^\circ = 0.174$.

Answer: 27

Solution:

$\triangle ABC$, by sine rule

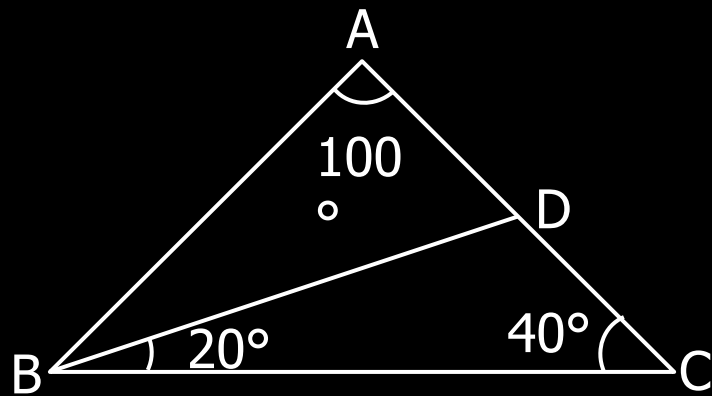
$$\frac{BC}{\sin 120^\circ} = \frac{BD}{\sin 40^\circ} = \frac{20}{\sin 40^\circ}$$

$$\therefore BC = \frac{10\sqrt{3}}{\sin 40^\circ}$$

Given, $\cos 80^\circ = (\sin 10^\circ) = 0.174$

$$\sin 40^\circ = \sqrt{\frac{1 - \cos 80^\circ}{2}} = 0.643$$

$$\therefore \text{Nearest integer to } BC = 27$$



$$720 - 24p = 24^2 + 30^2 + p^2 - 60p - p^2$$

SMO - 2013

Amazing Question!

Best Question
Today's Session

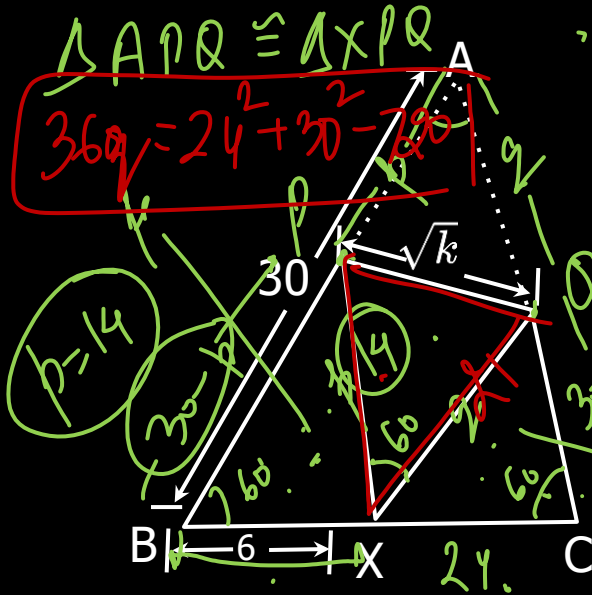
Q7. $\triangle ABC$ is an equilateral triangle of side length 30. Fold the triangle so that A touches a point X on BC. If $BX = 6$, find the value of k , where $\sqrt{k}$ is the length of the crease obtained from folding.

$$\cos 60 = \frac{1}{2} = \frac{24^2 + (30-p)^2 - p^2}{2 \cdot 24 \cdot (30-p)}$$

$$9p = 6 + 150 - 30$$

$$9p = 126$$

$$p = 14$$



$$\triangle BPX \sim \triangle CQX$$

$$\cos 60 = \frac{6^2 + (30-p)^2 - p^2}{2 \cdot 6 \cdot (30-p)}$$

$$2 = 21$$

$$6^2 + 30^2 + p^2 - 60p - p^2 = 180 - 6p$$

$$54p = 6^2 + 30^2 - 180$$

$$36p = 24^2 + 180$$

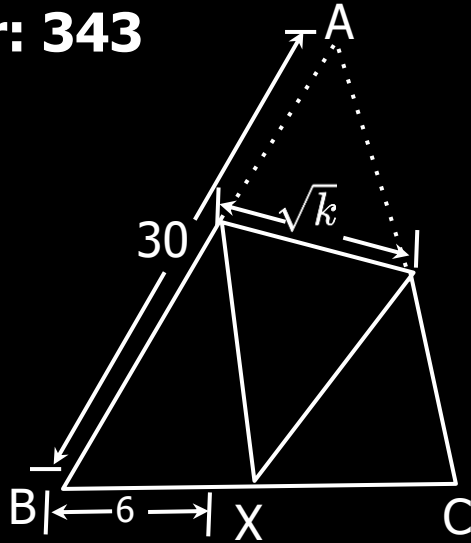
$$9p = 165 + 180$$

$$9p = 345$$

$$p = 38.33$$

Q7. $\triangle ABC$ is an equilateral triangle of side length 30. Fold the triangle so that A touches a point X on BC. If $BX = 6$, find the value of k , where $\sqrt{k}$ is the length of the crease obtained from folding.

Answer: 343



Pre RMO - 5 marks

Solution:

Apply the law of cosine on ΔXBY and ΔXCZ respectively:

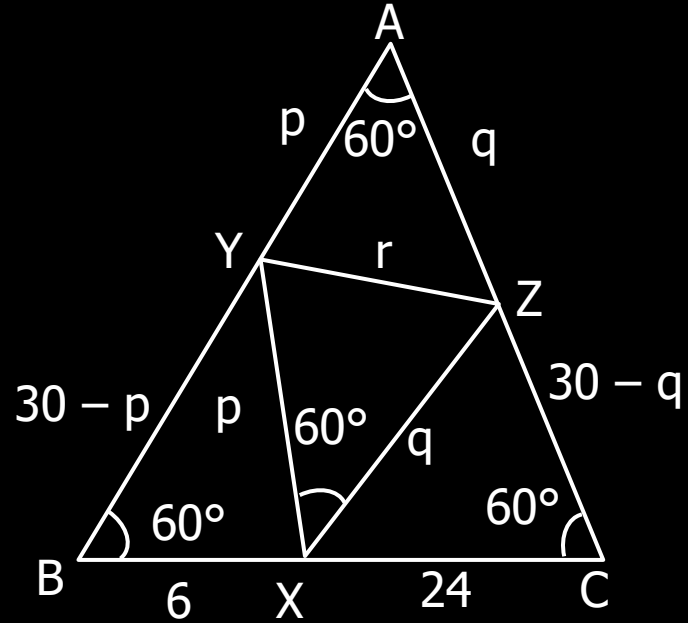
$$p^2 = 6^2 + (30 - p)^2 - 6(30 - p),$$

$$q^2 = 24^2 + (30 - q)^2 - 24(30 - q).$$

Then $p = 14$ and $q = 21$.

Applying the law of cosine in ΔXYZ again to obtain

$$k = r^2 = p^2 + q^2 - pq = 14^2 + 21^2 - 14 \cdot 21 = 343.$$



$$\theta > 45^\circ$$

RMO - 2007

Q8. Let ABC be an acute-angled triangle; AD be the bisector of $\angle BAC$ with D on BC ; and BE be the altitude from B on AC . Show that $\angle CED > 45^\circ$.

$$b(\sin \theta \cos C + \cos \theta \sin C) = b \sin \theta \cos C + c \sin \theta \cos C$$

$$b \cos \theta \sin C = c \sin \theta \cos C \Rightarrow \tan \theta = \frac{b \sin C}{c \cos C} = \frac{b}{c} \tan C$$

$$CE = a \cos C$$

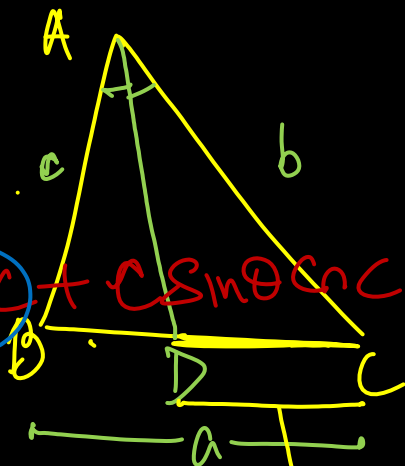
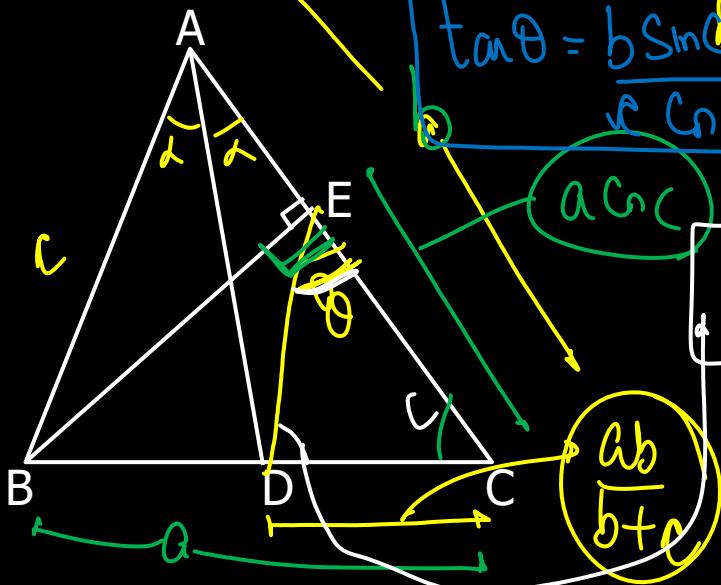
$$\tan \theta = \frac{b \sin C}{c \cos C}$$

$$\frac{CD}{\sin \theta} = \frac{CE}{\sin(180 - (C + \theta))}$$

$$\frac{ab}{b+c}$$

$$\frac{ab}{(b+c) \sin \theta} = \frac{a \cos C}{\sin(C + \theta)}$$

$$b \sin(C + \theta) = (b+c) \sin \theta \cos C$$



Solution:

Let $\angle CED = \theta$. We have $CD = \frac{ab}{b+c}$ and $CE = a \cos C$.
Using sine rule in triangle CED, We have

$$\frac{CD}{\sin \theta} = \frac{CE}{\sin(C+\theta)}.$$

This reduces to

$$(b+c) \sin \theta \cos C = b \sin C \cos \theta + b \cos C \sin \theta.$$

Simplification gives $c \sin \theta \cos C = b \sin C \cos \theta$ so that

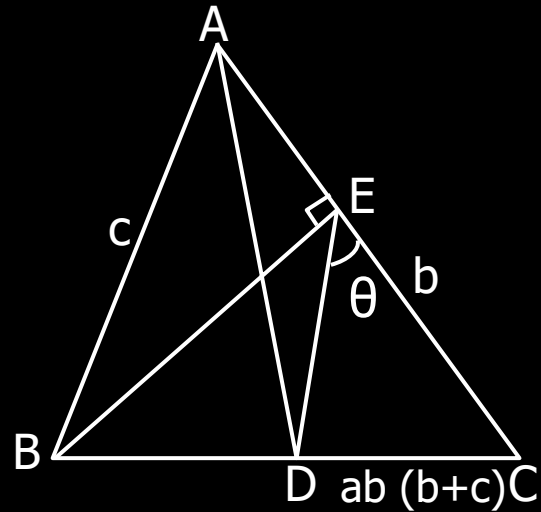
$$\tan \theta = \frac{b \sin C}{c \cos C} = \frac{\sin B}{\cos C} = \frac{\sin B}{\sin\left(\frac{\pi}{2} - C\right)}$$

Since ABC is acute-angled, we have $A < \frac{\pi}{2}$.

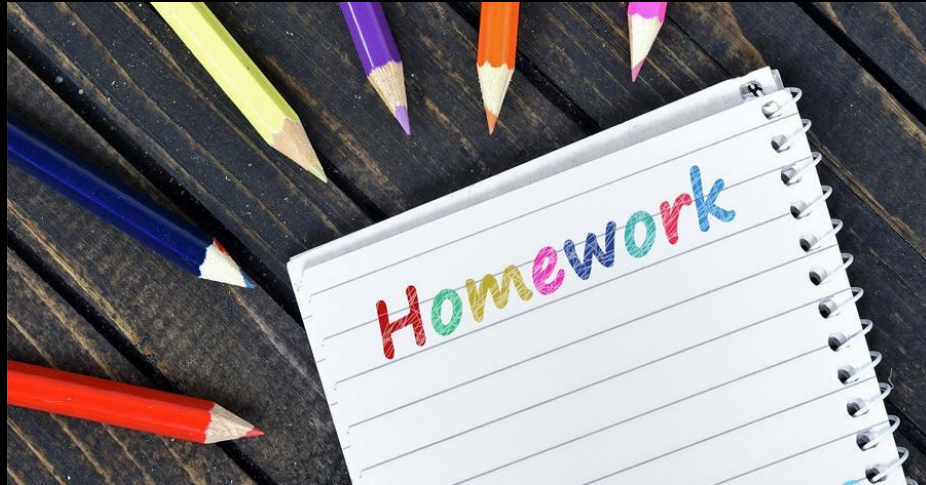
$$\text{Hence } B + C > \frac{\pi}{2} \text{ or } B > \left(\frac{\pi}{2}\right) - C.$$

$$\text{Therefore } \sin B > \sin\left(\frac{\pi}{2} - C\right).$$

$$\text{This implies that } \tan \theta > 1 \text{ and hence } \theta > \frac{\pi}{4}.$$

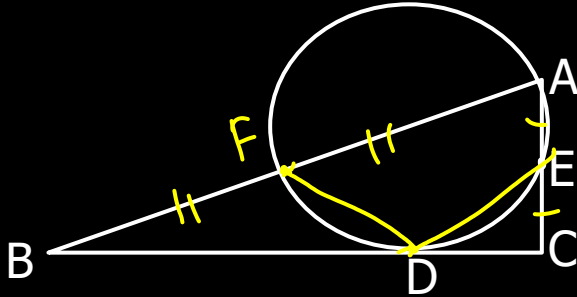


Home Work Sawaal



Q9. In the triangle ABC, a circle passes through the point A, the midpoint E of AC, the midpoint F of AB and is tangent to the side BC at D. Suppose $\frac{AB}{AC} + \frac{AC}{AB} = 4$. Determine $\angle EDF$ in degrees.

Trigo + Geometry





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abhay.mahajan@vedantu.com

ABHAY MAHAJAN
MATHEMATICS EXPERT

APRIL
MONTH SCHEDULE



Pre RMO

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ABHAY MAHAJAN
MATHEMATICS EXPERT

KVPY SA SCHEDULE – WEEK 1

**ABHAY
MAHAJAN**

Number Theory

**MONDAY
@ 11:00 AM**

**NISHANT
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**Kinematics and Laws of Motion:
Most Important Questions**

**WEDNESDAY
@ 12:30 PM**

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**Animal Tissue:
Part-1**

**WEDNESDAY
@ 4:00 PM**

**NIDHI
PRASHAR**

Atomic Structure

**FRIDAY
@ 11:30 AM**

KVPY SX SCHEDULE – WEEK 1

**ABHAY
MAHAJAN**

**Sequence and Series
and Quadratic Equations**

**WEDNESDAY
@ 11:00 AM**

**NISHANT
SHARMA**

**Kinematics and Laws of Motion:
Most Important Questions**

**THURSDAY
@ 12:30 PM**

**ASIF
TYAGI**

**Animal Reproduction:
Part-1**

**FRIDAY
@ 4:00 PM**

**NIDHI
PRASHAR**

Solutions and Colligative Properties

**SATURDAY
@ 11:30 AM**

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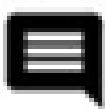
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Q9. In the triangle ABC, a circle passes through the point A, the midpoint E of AC, the midpoint F of AB and is tangent to the side BC at D. Suppose $\frac{AB}{AC} + \frac{AC}{AB} = 4$. Determine $\angle EDF$ in degrees.

Answer: 120°

Solution:

Let $BC = a$, $CA = b$ and $AB = c$. Let $BD = a_1$ and $DC = a_2$.

Using the power of B with respect to the circle,

$$\text{we have } a_2^2 = \frac{b^2}{2} \quad \text{Similarly } a_1^2 = \frac{c^2}{2}$$

$$\text{Thus } b + c = \sqrt{2}(a_1 + a_2)$$

$$= \sqrt{2}a, \text{ or } 2a^2 = (b + c)^2,$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{2b^2 + 2c^2 - (b + c)^2}{4bc}$$

$$= \frac{1}{4} \left(\frac{b}{c} + \frac{c}{b} - 2 \right) = \frac{1}{4} (4 - 2) = \frac{1}{2}$$

Therefore, $\angle A = 60^\circ$. Since A, F, D, E are concyclic, $\angle EDF = 120^\circ$

